

CashSnap Dataset Strategy

Fact-Checked Combined Report, Dataset Blueprint, Synthetic Data Plan, and Validation Gates

Prepared from the uploaded CashSnap reports and current public research sources. Date: 2026-06-06.

1. Executive Verdict

The strongest version of the uploaded reports is not: "make the dataset as large and diverse as possible." The better conclusion is: build a scoreable, bias-controlled real stress benchmark first, then use synthetic data as an experiment generator that is promoted only when it improves real fan, overlap, hand-occlusion, and partial-note counting across seeds and controls.

For CashSnap, a near-perfect dataset is defined by measured real-world transfer, not by a fixed image count. The dataset is good enough when a lightweight mobile/browser model plus fragment-fusion logic reaches stable physical-note count accuracy on real P1 partial/fan/overlap cases without raising hallucination on ambiguous slivers or hard negatives.

- **Best-supported direction:** Keep the lightweight detector for clean notes, add a visible-fragment/evidence path, and evaluate the fused physical-note count rather than summing fragments.
- **Main correction:** Several source reports overstate amodal labeling, synthetic-real ratios, and single-paper results. Amodal geometry can help, but exact visible evidence and parent IDs are more important than forcing every human annotator to draw perfect amodal masks.
- **Main missing asset:** A real, scoreable P1 fan/overlap/partial benchmark with fragment labels, physical parent IDs, and human-identifiability tags.
- **Main synthetic rule:** Scale WebGL only after small, matched, multi-seed experiments prove transfer to the real P1 stress set. Synthetic volume alone is not evidence.
- **Main currency correction:** The class taxonomy must be updated from official currency sources: USD has seven denominations currently issued; KHR in circulation includes 50 through 200,000 riels, including the 200,000 riel note issued in 2024 [16,17].

2. Fact-Check and Debiasing Audit of the Source Reports

The reports are directionally useful, but they mix established findings, engineering heuristics, speculative research, and a few unsupported or misapplied claims. The table below shows what was retained, corrected, or removed.

Claim in source reports	Verdict	Correction used in this report
Dataset diversity matters more than raw volume.	Supported, but incomplete.	Keep it, but define diversity along CashSnap axes: denomination, side, design/series, wear, visibility fraction, overlap topology, hand/finger occlusion, repeated same-class bills, phone/camera conditions, backgrounds, and hard negatives.
Domain randomization and structured domain randomization are useful for sim-to-real.	Supported.	Use a structured scene grammar: realistic bills on realistic retail surfaces with randomized lighting, camera, wear, hands, and clutter. Do not render bills floating in arbitrary 3D space [1,2].
Copy-paste augmentation is powerful.	Supported.	Use copy-paste as a baseline and for rare classes, but QA edge seams, scale, lighting, and context. Context-free paste can hurt detection [3,4,5].
Synthetic data can reduce the amount of real data needed by up to 70% and 5-20% real data can be effective.	Supported only as a specific published finding, not a CashSnap rule.	Burdorf et al. found this in a specific object-detection study [9]. Treat 5-20% real as a candidate sweep, not the target ratio. CashSnap needs its own P1 real stress proof.
Golden Ratio Mixing is an optimal real/synthetic ratio.	Misapplied.	The golden-ratio paper concerns stabilizing recursive generative-model training under theoretical settings, not choosing an object-detection synthetic-real ratio for banknotes [20]. Exclude it from CashSnap training decisions.
Every instance must have amodal polygons/masks.	Overclaim.	For CashSnap, exact visible masks/boxes plus physical parent IDs are mandatory. Amodal quadrilaterals/boxes are useful when annotators can infer note geometry. Full amodal masks should be optional or synthetic-only because they are expensive and can encourage hallucination [11,12].
OBB is the main solution for hard fans/overlaps.	Incorrect for CashSnap.	Use OBB only when the visible region is compact and honestly rectangle-like. For fans, stacks, cut-off notes, hand splits, and slivers, use visible polygons/masks and fragment-parent fusion.
Banknote occlusion recognition at 98.07% under 75% occlusion.	Not verified from accessible primary sources.	Do not rely on this result. Retain verified banknote evidence: component-based recognition handles occlusion, clutter, viewpoint changes, and worn notes; Costa et al. specifically evaluated partially visible, folded, wrinkled/worn, and overlapping Euro notes [13,14].

YOLOv11 mixed-currency paper proves the CashSnap solution.	Useful but not proof.	Recent mixed/partly-covered currency detection work is relevant, but CashSnap is harder because it requires physical-note counting from severe overlap and fragment fusion on USD+KHR in mobile/browser deployment [15].
More real data is always better.	False for this project history.	Your history shows background-heavy all-real curricula and duplicate rare oversampling can collapse or fail. The dataset must be balanced by exposure, order, condition, and validation controls, not just image count.
Dataset closeness metrics such as FID/MMD can validate synthetic data.	Diagnostic only.	Use feature-space separability, label-distribution matching, heatmaps, and coverage dashboards to find gaps, but never promote synthetic data based on closeness metrics alone. Real P1 transfer is the gate.
Counterfeit detection should be included.	Out of scope.	Keep the report focused on denomination/counting. Use lawful training imagery and avoid print-ready counterfeit-grade note generation; the generator should be for CV training, not authenticity reproduction.

3. Research Findings That Should Drive CashSnap

3.1 Dataset completeness means coverage of target conditions, not abstract bigness

A complete CashSnap dataset must cover the deployment distribution at the level where failures occur: note class, note side/design, visible fraction, overlap topology, repeated-note topology, camera profile, lighting, background, hand/finger occlusion, and hard-negative clutter. ObjectNet is a cautionary example: when background, rotation, and viewpoint are controlled rather than left biased, models can show large performance drops [10].

The right unit of "enough data" is not only images per class. It is class-by-condition exposure and held-out stress performance. For example, KHR 50,000 under same-denomination fan with 10-25% visibility is a different risk cell than KHR 50,000 fully visible on a clean counter.

3.2 Synthetic data is valuable, but only when it wins on real stress tests

The synthetic data literature supports using simulation, domain randomization, structured domain randomization, copy-paste, and refinement. It does not support blind scaling. Tobin et al. showed that broad randomization can transfer from simulation to real objects [1]. Prakash et al. showed that context-aware structured domain randomization can outperform naive randomization [2]. Dwibedi and Ghiasi showed that cut-paste can be surprisingly effective [3,5]. Dvornik showed that context matters and poor placement can hurt [4].

For CashSnap, the correct use of WebGL is to generate controlled experiments: "does adding this occlusion topology, rare KHR class, camera blur distribution, or hand mask improve real P1 transfer across seeds?" If not, the generator is producing plausible pictures but not useful training evidence.

3.3 Occluded banknote counting is not normal object detection

Standard detection labels can double-count one physical note when a hand splits it into multiple visible fragments, and can undercount when only thin edges or denomination cues remain. CashSnap needs two evaluation targets: visible-evidence detection and physical-note count after fragment fusion. Amodal labels can support fusion, but they must not force the model to hallucinate a denomination when the visible fragment is below the human-identifiable threshold.

3.4 Banknote-specific evidence favors local cues and component supervision

Banknote recognition papers show that local components such as portraits, seals, numerals, text blocks, colors, and geometric validation can be robust under occlusion, clutter, viewpoint change, and worn notes [13,14]. This supports CashSnap's fragment/evidence path: supervise local denomination cues instead of expecting a small detector to infer whole notes from partial boxes alone.

4. Practical Definition of the Perfect CashSnap Dataset

A perfect CashSnap dataset is a dataset-plus-validation system that makes the following statement true:

"When a human can identify and count the physical notes from one casual retail photo, the model can do so with stable accuracy across cameras, backgrounds, occlusion patterns, and random training seeds; when a fragment is not human-identifiable, the model abstains rather than inventing a denomination."

- **Coverage:** Every target denomination and side/design appears across the high-risk conditions, especially partial/fan/overlap and repeated same-class layouts.

- **Honest labels:** Visible evidence is labeled exactly; ambiguous fragments are marked unknown/ignore; fragments are linked to physical-note parent IDs.
- **Fusion supervision:** Labels allow evaluation of both fragment detection and physical-note count so a note split into three visible pieces is still one note.
- **Bias control:** Train, validation, and test splits are separated by session, camera, background, and layout recipe, not random image rows.
- **Validated synthetic:** Synthetic data is accepted only when it improves real P1 stress performance against matched controls and seed-stability gates.
- **Deployable:** The dataset supports the model size, latency, browser/mobile memory, and quantization constraints of CashSnap, not just offline mAP.

5. Recommended CashSnap Dataset Blueprint

5.1 Class taxonomy

Use a current, explicit taxonomy. Do not let missing or outdated denominations hide failure modes.

- **USD:** \$1, \$2, \$5, \$10, \$20, \$50, \$100. The \$2 class may be lower priority operationally, but it is a currently issued denomination and should be in the taxonomy [16].
- **KHR:** 50, 100, 200, 500, 1,000, 2,000, 5,000, 10,000, 15,000, 20,000, 30,000, 50,000, 100,000, and 200,000 riels. The National Bank of Cambodia lists the 200,000 riel note issued on 16 October 2024 [17].
- **Design/series:** Treat older/current designs as metadata initially. Promote to subclasses only if the model confuses designs or if deployment needs design-specific counting.
- **Side:** Keep side as a required label field. For the fragment classifier, side-specific cue learning is useful because portraits, numerals, and text differ by face/back.

5.2 Minimum and target image plan

The numbers below are deliberately framed as decision thresholds, not universal laws. Increase counts in the error cells that remain weak on real P1 tests.

Dataset part	Recommended volume	Purpose and caveat
Real P1 stress benchmark	Minimum 1,200-2,000 images before synthetic promotion. Target 3,000-5,000.	Held out. Must include fan, overlap, stack, hand/finger, border truncation, repeated same-class, sliver/unknown, hard negatives. Score physical counts and total value.
Real training set	Minimum 5,000-8,000 diverse labeled images. Target 15,000-30,000 over time.	Split by collection session, camera, background, and operator. Avoid background-heavy row inflation.
Synthetic development batches	Start with 25,000-50,000 images per recipe family.	Small enough for dose tests; large enough to expose class/condition effects.
Synthetic promoted pool	After gates, 100,000-250,000. Later 500,000-1,000,000 only if P1 transfer remains stable.	Allocate synthetics to failure cells, not uniformly to every class.
Per-class exposure heuristic	Aim for at least 1,500 useful images/class and around 10,000 total instances/class across real plus validated synthetic.	Use as a planning heuristic from detector training practice, not as a proof of sufficiency [19].
Rare/high-risk cells	At least 150-300 real validation instances per high-risk class-condition cell where practical; supplement training with synthetic.	Especially KHR 50k/100k/200k and any KHR denominations that collapse under training.

5.3 Required coverage axes

- **Denomination and side:** USD and KHR, front/back, series/design metadata, and repeated same-denomination layouts.
- **Physical condition:** crisp, circulated, worn, dirty, creased, folded, curled, damp-looking, stained, torn edge, faded ink.
- **Visibility bins:** >75%, 50-75%, 25-50%, 10-25%, and <10% sliver. The last bin should often be unknown/ignore unless human-identifiable cues exist.
- **Occlusion topologies:** single clean note, pair overlap, 3-6 note fan, 3-6 repeated same-denomination fan, mixed-currency fan, stacked pile, hand/finger cover, strap/receipt occlusion, border-truncated note, disjoint fragments of one note.
- **Camera and imaging:** top-down and oblique angles, 0.5x/1x/2x phone focal lengths, close and far distances, focus blur, motion blur, compression, exposure shifts, glare, shadows, lens dirt, white-balance variation.
- **Backgrounds:** retail counters, mats, cloth, wallets, bags, receipts, cards, coins, patterned paper, colored paper, clutter, and no-bill hard negatives.
- **Scene composition:** 1-2 note clean scenes, 3-6 note counting scenes, 7-15 note high-density stacks/fans, mixed USD/KHR, and total-value edge cases where one missed high-value bill dominates error.

5.4 Split design

- **Train:** May include real plus promoted synthetic. Do not leak near-duplicate backgrounds, camera bursts, or synthetic recipes into validation/test.
- **Validation:** Real-only primary validation plus a small synthetic sanity split for generator QA. Tune thresholds on validation, not test.
- **Test A - Clean ID:** Mostly visible notes in normal retail scenes, to protect the simple detector path.
- **Test B - Retail ID:** Casual real photos with ordinary clutter, hands, and mixed currencies.
- **Test C - P1 stress:** Heavy partial/fan/overlap/stack/hand/border cases with physical parent IDs. This is the main promotion gate.
- **Test D - Hard negative/unknown:** No bills, receipts/cards/paper, and slivers that humans cannot identify. Measures hallucination and abstention.
- **Test E - OOD:** New phone models, backgrounds, stores, lighting, and collection sessions not seen in training.

6. Label Schema and Annotation Rules

The label design must make the fusion problem scoreable. The label does not exist only to train YOLO boxes; it also exists to prove whether the post-processor counted physical notes correctly.

Field	Definition
image_id	Unique image identifier.
source	real, synthetic, copy_paste, webgl, or augmented_real.
session_id / camera_id	Used to prevent split leakage.
scene_layout_type	single, pair_overlap, fan_same, fan_mixed, stack, hand_occlusion, border_partial, hard_negative, etc.
physical_note_id	Stable parent ID for one physical bill.
fragment_id	Visible fragment ID. One physical note may have multiple fragments.
currency / denomination / side / design_series	Class fields and visual design fields.
visible_polygon or mask	Exact visible pixels or polygon of the fragment. This is the authoritative detection target.
visible_bbox	Axis-aligned box derived from visible mask.
oriented_bbox	Only if the visible fragment is compact and rectangle-like; otherwise omit.
amodal_quad or amodal_box	Optional human-inferable full-note geometry; useful for fusion. Do not require full amodal masks for every real label.
visibility_fraction	visible_mask_area / expected_note_area, measured or estimated.
occluder_ids / occlusion_order	Which object is on top when known, especially in synthetic and controlled real scenes.
human_identifiable	true, false, or uncertain. false/uncertain should train abstention/ignore, not forced denomination.
cue_types_visible	e.g., numeral, portrait, text, border pattern, color field, seal, serial-area, size/edge only.
ignore_reason	ambiguous_sliver, motion_blur, label_uncertain, counterfeit/out_of_scope, duplicate_uncertain, etc.
qa_status	unreviewed, first_pass, adjudicated, rejected.

6.1 Annotation decision rules

- **Label visible evidence first:** Draw the visible mask/polygon before attempting any full-note geometry.
- **Do not guess denominations:** If the annotator cannot identify the denomination from visible evidence, mark unknown/ignore even if a generator or scene script knows the hidden class.
- **Link fragments:** If one bill is visible in two or more disconnected areas, each fragment gets its own fragment_id and the same physical_note_id.
- **Amodal geometry is helper data:** Draw an amodal quad/box when the physical note footprint is reasonably inferable. Do not force complex amodal masks in real data.
- **OBB is conditional:** Only draw OBB when the visible region is compact and honest; do not wrap a fan, separated fragments, or hand-split note in one oriented rectangle.
- **Label hard negatives:** Include negative images and ambiguous currency-like objects so the abstention path has training and evaluation data.

7. Synthetic Data Generation Blueprint

The generator should be treated as a controlled experiment engine. Its job is to fill known gaps, not to produce infinite attractive images.

7.1 Generator design

- **Lawful note assets:** Use approved training imagery/scans and comply with currency-image rules. Avoid print-ready counterfeit-grade assets; the generator should be downsampled/watermarked or otherwise constrained for CV training only.
- **Scene grammar:** Sample realistic note layouts on retail surfaces: singles, fans, stacks, mixed-currency piles, repeated same-class fans, notes partially out of frame, notes held by hands, and notes occluded by receipts/cards/fingers.
- **Two synthetic lanes:** Use about 70% structured realistic randomization and 30% stronger domain randomization as a starting sweep. Tune based on real P1 transfer, not aesthetics.
- **Renderer ground truth:** Output visible masks, physical parent IDs, amodal quads, occlusion order, visibility fractions, cue visibility, blur/exposure metadata, and generator recipe ID.
- **Hands and occluders:** Use real hand masks and synthetic hands, with plausible contact, shadows, and scale. Finger occlusion is not generic clutter; it is a top CashSnap failure mode.
- **Wear and material variation:** Creases, folds, curled edges, dirt, damp-looking patches, stains, faded ink, torn corners, and specular highlights should be independently parameterized.
- **Camera pipeline:** Randomize focal length, crop, resolution, JPEG artifacts, motion/focus blur, exposure, white balance, vignetting, lens dirt, and rolling-shadow effects.
- **Background library:** Use real retail/table textures plus hard-negative clutter. Decouple each denomination from any single background, camera, or layout recipe.

7.2 Synthetic QA gates

- **No ghost labels:** Every positive training label must have visible pixels unless explicitly used for amodal-only auxiliary training. Completely hidden bills must not train denomination detection.
- **Visibility calibration:** Check the empirical distribution of visibility bins by class and topology. High-risk cells should be intentionally oversampled.
- **Artifact audit:** Train a synthetic-vs-real classifier on embeddings. High separability is a warning sign, not an automatic rejection.
- **Context audit:** Flag impossible placements: floating bills, impossible shadows, implausible hand scale, inconsistent lighting, repeated texture clones, and hard-edged cut-paste seams.
- **Human review:** Manually review random samples from each generator recipe and every high-risk rare-class recipe before training.
- **Dose logging:** Every synthetic batch must have recipe IDs, class/condition counts, generation seed, row count, ordering, and augmentation settings so experiments are reproducible.

8. Validation Protocol and Promotion Gates

This protocol is designed around CashSnap's actual failure history: row-count/order sensitivity, seed instability, real-data curriculum collapse, duplicate oversampling failure, and missing scoreable P1 transfer proof.

8.1 Required experiment matrix

Experiment	What it proves
Clean baseline	Train the current lightweight detector on clean/mostly visible notes. Protect this path from regressions.
Real-only capped controls	Use capped, balanced real subsets to check whether synthetic beats real exposure rather than row count.
Synthetic-only control	Measures whether generator labels and textures are learnable, but is not deployable proof.
Mixed real+synthetic	Test multiple ratios and schedules: 5%, 10%, 20%, 50% real; mix during training vs synthetic pretrain + real fine-tune.
Matched row-count control	Same number of rows as the synthetic recipe but filled with neutral/duplicate/real controls.
Matched class-exposure control	Same class counts and object counts so gains cannot be explained by rare-class dose only.
Fixed-step/order-stable control	Same number of optimizer steps and shuffled order rules across variants.
Rare-class ablation	Compare rare KHR synthetic, duplicate-only oversampling, real rare crops, and copy-paste rare crops.
Fusion ablation	Evaluate detector-only, fragment-only, and detector+fusion to prove that fusion reduces overcount/undercount.
Generator ablation	Add one axis at a time: hands, fanned stacks, wear, camera blur, backgrounds, glare, hard negatives.

8.2 Metrics

- **Visible-fragment detection:** mAP50-95, mAP50, recall at fixed precision, AP by denomination, AP by visibility bin, AP by topology.
- **Fragment classification:** denomination accuracy on human-identifiable fragments; top-2 accuracy only as diagnostic; unknown false-positive rate.
- **Physical-note counting:** exact count accuracy, count MAE, overcount rate, undercount rate, duplicate-fragment overcount rate, missed-note undercount rate.
- **Total value:** absolute value error and worst-case high-value error by currency.
- **Abstention/hallucination:** false denomination rate on non-identifiable slivers, no-bill images, receipts/cards, and colored paper.
- **Fusion metrics:** fragment-to-parent precision/recall, split-note merge success, two-note non-merge success.
- **Stability:** mean and confidence intervals over at least 3 seeds; 5 seeds for promotion-critical changes.
- **Deployability:** browser/mobile latency, memory, model size, quantized accuracy, and failure modes under reduced resolution.

8.3 Promotion gates for synthetic data

Promote a synthetic recipe into the main training pool only when all of the following are true against matched controls:

1. P1 real stress exact-count accuracy improves by at least 3 percentage points, or P1 total-value MAE drops by at least 10%, without hiding failures in a class average.
2. No high-value or rare class loses more than 2 AP points or 2 percentage points in exact-count accuracy unless the change is explicitly accepted.
3. Unknown/sliver/hard-negative hallucination does not increase materially; use a starting guardrail of no more than 0.5-1.0 percentage point increase.
4. At least 3 independent seeds are positive; for large promotions prefer 5 seeds and require the confidence interval not to include a practically harmful effect.
5. Matched row-count and matched class-exposure controls do not explain the gain.
6. Mobile/browser model size and latency remain within deployment budget after quantization.
7. Qualitative error review shows plausible fixes, not new shortcut learning such as background or synthetic seam reliance.

9. Prioritized Action List

8. **Build the P1 real stress benchmark first.** Collect and label 1,200-2,000 real fan/overlap/partial/hand/stack/border images with visible masks, parent physical_note_id, human_identifiable, and total-value labels. This is the missing proof asset.
9. **Fix the currency taxonomy.** Update USD and KHR classes using official sources, including KHR 200,000 and low-value KHR notes. Decide which KHR designs are metadata vs subclasses.
10. **Write the annotation guide.** Document exactly when to label visible fragments, unknown/ignore, amodal quad, OBB, and parent IDs. Add examples of ambiguous slivers.
11. **Build WebGL Generator v0 with QA exports.** Before scale, output masks, parent IDs, occlusion order, visibility fraction, recipe IDs, and automated no-ghost-label checks.
12. **Run dose sweeps, not one-off wins.** Test small synthetic doses with fixed row-count/order controls and 3 seeds before any full-scale render.
13. **Add hard negatives early.** Receipts, cards, colored paper, wallets, paper stacks, no-bill clutter, and non-identifiable slivers should be first-class evaluation items.
14. **Focus rare-class rendering on failure cells.** Do not just duplicate rare KHR notes. Generate rare KHR under the same occlusion, fan, camera, and background cells where real validation fails.
15. **Add a fusion benchmark.** Score whether one physical note split into multiple fragments is counted once, and whether two overlapping identical notes are counted twice.
16. **Scale synthetic only after promotion gates.** If a recipe passes matched controls across seeds, expand it. If it passes one seed only, keep it in research, not production training.
17. **Keep a regression dashboard.** Every training run should show clean path AP, P1 count, rare-class performance, unknown hallucination, and mobile latency side by side.

10. Bottom-Line Blueprint

CashSnap should become a fragment-aware counting system trained on a hybrid dataset with three pillars:

- **Real stress truth:** A held-out, human-verified P1 set is the source of truth for whether the system works.

- **Structured synthetic scale:** WebGL should generate missing condition cells with exact masks and parent IDs, not generic pretty images.
- **Promotion science:** Every data addition must beat matched row-count and matched class-exposure controls across seeds and must not increase hallucination.

The ideal first milestone is not a million-image synthetic dataset. It is a 1,200-2,000 image real P1 stress benchmark, a corrected USD/KHR taxonomy, a mask/parent-ID label schema, and a WebGL batch that demonstrably improves P1 physical-note counts over controls. Once that is proven, scale the renderer aggressively into the error cells.

Appendix A. Papers and Findings Table

This table keeps the useful research from the source reports, removes unsupported extrapolations, and adds CashSnap-specific actions.

Paper	Year	Topic	Dataset type	Relevant conditions	Key result	What CashSnap should copy	What CashSnap should avoid
Tobin et al., Domain Randomization	2017	Sim-to-real domain randomization	Synthetic simulator	Synthetic, random textures, distractors, partial occlusion	With enough randomized rendering, real scenes can become another variation; demonstrated real object localization from synthetic training [1].	Use heavy variation in texture, light, camera, and distractors.	Do not rely on photorealism alone or easy clean views.
Prakash et al., Structured Domain Randomization	2019	Context-aware synthetic data	Synthetic contextual scenes	Synthetic, context, sim-to-real	Contextually plausible object placement improves over naive random placement [2].	Use retail scene grammar: notes on counters, in hands, near receipts/cards.	Do not float bills randomly or sample impossible layouts.
Dwibedi et al., Cut, Paste and Learn	2017	Cut-paste detection data	Synthetic composites + real	Copy-paste, synthetic, rare contexts	Patch-level realism can be enough; synthetic+real improved detection, including low-real regimes [3].	Paste bill masks/fragments into varied real backgrounds as baseline.	Do not assume edge artifacts are harmless; audit seams.
Dvornik et al., Modeling Visual Context	2018	Context-aware copy-paste	Detection augmentation	Copy-paste, context, few labels	Adding objects can hurt if context is wrong; context modeling improves mAP when labels are few [4].	Match bill scale, surface, lighting, and plausible placement.	Do not paste bills into impossible or misleading contexts.
Ghiassi et al., Simple Copy-Paste	2021	Copy-paste for instance segmentation	COCO/LVIS augmentation	Copy-paste, rare categories	Simple random copy-paste is strong and improved rare LVIS categories [5].	Use simple copy-paste as a strong low-complexity baseline.	Do not over-engineer before beating this baseline.
Shrivastava et al., SimGAN	2017	Adversarial synthetic refinement	Synthetic + unlabeled real	Synthetic-real gap	Refines simulated images while preserving labels using adversarial training [6].	Consider only if renderer artifacts are persistent and measured harmful.	Do not add GAN refinement without label-preservation QA.
Mayer et al., Good Synthetic Data	2018	Synthetic data properties and schedules	Optical flow/disparity synthetic sets	Synthetic, curriculum/scheduling	Training schedules and synthetic data properties strongly affect transfer [7].	Treat curriculum/order as experiment variables.	Do not dump all data into one untracked mix.
Nowruzi et al., How Much Real Data?	2019	Synthetic/real object detection analysis	Mixed real/synthetic	Mixing, domain gap	Analyzes replacing real data with synthetic and domain similarity for object detection [8].	Run CashSnap-specific real-only, synth-only, mixed, and fine-tune controls.	Do not assume published ratios transfer directly.
Burdorf et al., Reducing Real Data with Synthetic	2022	Synthetic-real mix and scaling	Mixed object detection	Synthetic, real savings, class imbalance	Found up to 70% real-data reduction and effective 5-20% real mixes in their setting [9].	Use 5-20% real as one sweep range, especially for rare cells.	Do not treat 5-20% or 70% savings as guaranteed.
Barbu et al., ObjectNet	2019	Bias-controlled evaluation	Real test set	Background, rotation, viewpoint bias	Controlling nuisance variables exposed large model drops [10].	Build CashSnap OOD and P1 benchmarks with controlled backgrounds/cameras.	Do not trust random validation splits from same sessions.
Hsieh et al., TAO-Amodal	2023/2024	Modal + amodal tracking labels	Real videos	Amodal boxes, occlusion, visibility	Provides modal/amodal boxes and visibility for many categories [11].	Use modal visible labels plus optional amodal boxes/quads.	Do not force full amodal masks when annotators cannot infer them.
Zhu et al., Semantic Amodal Segmentation	2017	Amodal segmentation	Amodal segmentation benchmark	Occlusion, modal/amodal masks	Shows value of distinguishing visible and full object extent [12].	Use amodal geometry as auxiliary fusion signal.	Do not let amodal training override visible evidence.
CountOCC / amodal counting preprints	2025+	Counting through occlusion	Research/preprint	Occlusion, amodal counting	Recent work explores reconstructing occluded features for counting, but maturity and transfer to banknotes are unproven.	Borrow the idea of evaluating occlusion-specific counting errors.	Do not build around speculative heavy architecture before labels exist.
Hasanuzzaman et al., Component-based Banknote Recognition	2012	Banknote recognition for blind users	Real banknote images	Occlusion, clutter, viewpoint, worn notes	Component/SURF features address occlusion, clutter, viewpoint, and worn/wrinkled bills [13].	Supervise local denomination cues: portraits, numerals, seals, borders.	Do not rely only on whole-note templates.
Costa et al., Banknotes in Multiple Perspectives	2015/2016	Feature matching and shape validation	Euro banknote tests	Partial, folded, worn, overlap	Reported tests with clutter, perspective, folding, partial occlusion, and overlapping banknotes [14].	Use geometry validation and local cues for fragment evidence.	Do not assume one global detector solves folds/partial views.
Fanzury and Hwang, YOLOv11 Mixed/Partly Covered Currency	2025	Mobile mixed-currency detection	Purpose-built real dataset	Mixed currency, partial cover, mobile	Relevant mobile/object-detection reference for mixed and partly covered banknotes [15].	Use as a mobile deployment reference, not a complete counting solution.	Do not confuse detection under partial cover with physical-note counting under severe fan/overlap.
U.S. Currency Education Program	current	Official USD denominations	Official public source	Class taxonomy	Lists the seven USD denominations: \$1, \$2, \$5, \$10, \$20, \$50, \$100 [16].	Include \$2 as a valid but possibly low-priority class.	Do not omit valid classes without explicit product decision.
National Bank of Cambodia	current	Official KHR in-circulation banknotes	Official public source	Class taxonomy, KHR designs	Lists KHR denominations including 200,000 riels issued in 2024 [17].	Update KHR taxonomy and design metadata.	Do not use outdated KHR class lists.
Ultralytics YOLO training guidance / YOLO26 docs	current	Detector engineering heuristics	Engineering docs	Small/mobile object detection	Provides practical dataset heuristics and current lightweight detector options [18,19].	Use heuristics for planning and latency guardrails.	Do not cite heuristics as scientific laws.

References

- [1] Tobin et al. 2017. Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World. arXiv:1703.06907. <https://arxiv.org/abs/1703.06907>
- [2] Prakash et al. 2019. Structured Domain Randomization: Bridging the Reality Gap by Context-Aware Synthetic Data. arXiv:1810.10093 / ICRA. <https://arxiv.org/abs/1810.10093>
- [3] Dwibedi et al. 2017. Cut, Paste and Learn: Surprisingly Easy Synthesis for Instance Detection. arXiv:1708.01642 / ICCV. <https://arxiv.org/abs/1708.01642>
- [4] Dvornik, Mairal, Schmid. 2018. Modeling Visual Context is Key to Augmenting Object Detection Datasets. ECCV. https://openaccess.thecvf.com/content_ECCV_2018/html/NIKITA_DVORNIK_Modeling_Visual_Context_ECCV_2018_paper.html
- [5] Ghiasi et al. 2021. Simple Copy-Paste is a Strong Data Augmentation Method for Instance Segmentation. arXiv:2012.07177 / CVPR. <https://arxiv.org/abs/2012.07177>
- [6] Shrivastava et al. 2017. Learning from Simulated and Unsupervised Images through Adversarial Training. arXiv:1612.07828 / CVPR. <https://arxiv.org/abs/1612.07828>
- [7] Mayer et al. 2018. What Makes Good Synthetic Training Data for Learning Disparity and Optical Flow Estimation? arXiv:1801.06397 / IJCV. <https://arxiv.org/abs/1801.06397>
- [8] Nowruzi et al. 2019. How much real data do we actually need: Analyzing object detection performance using synthetic and real data. arXiv:1907.07061. <https://arxiv.org/abs/1907.07061>
- [9] Burdorf, Plum, Hasenklever. 2022. Reducing the Amount of Real World Data for Object Detector Training with Synthetic Data. arXiv:2202.00632. <https://arxiv.org/abs/2202.00632>
- [10] Barbu et al. 2019. ObjectNet: A large-scale bias-controlled dataset for pushing the limits of object recognition models. NeurIPS. <https://objectnet.dev/> and <https://openreview.net/forum?id=SkxJ-309FQ>
- [11] Hsieh et al. 2023/2024. TAO-Amodal: A Benchmark for Tracking Any Object Amodally. <https://tao-amodal.github.io/>
- [12] Zhu, Tian, Metaxas, Dollár. 2017. Semantic Amodal Segmentation. CVPR. https://openaccess.thecvf.com/content_cvpr_2017/html/Zhu_Semantic_Amodal_Segmentation_CVPR_2017_paper.html
- [13] Hasanuzzaman, Yang, Tian. 2012. Robust and Effective Component-based Banknote Recognition for the Blind. IEEE Transactions on Systems, Man, and Cybernetics. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3363999/>
- [14] Costa et al. 2015/2016. Recognition of Banknotes in Multiple Perspectives Using Selective Feature Matching and Shape Analysis. https://www.researchgate.net/publication/286454339_Recognition_of_Banknotes_in_Multiple_Perspectives_Using_Selective_Feature_Matching_and_Shape_Analysis
- [15] Fanzury and Hwang. 2025. Real-Time Identification of Mixed and Partly Covered Foreign Currency Using YOLOv11 Object Detection. MDPI Sensors. <https://www.mdpi.com/1424-8220/25/11/3518>
- [16] U.S. Currency Education Program. The Seven Denominations. <https://www.uscurrency.gov/denominations>
- [17] National Bank of Cambodia. Banknotes in Circulation. https://www.nbc.gov.kh/english/about_the_bank/banknotes_in_circulation.php
- [18] Ultralytics. YOLO26 documentation and model information. <https://docs.ultralytics.com/models/yolo26/>
- [19] Ultralytics. YOLO training tips / dataset heuristics. https://docs.ultralytics.com/yolov5/tutorials/tips_for_best_training_results/
- [20] He et al. 2025. Golden Ratio Mixing of Real and Synthetic Data for Stabilizing Generative Model Training. arXiv:2506.00712. <https://arxiv.org/abs/2506.00712>
- [21] Beery et al. 2020. Synthetic Examples Improve Generalization for Rare Classes. WACV. https://openaccess.thecvf.com/content_WACV_2020/html/Beery_Synthetic_Examples_Improve_Generalization_for_Rare_Classes_WACV_2020_paper.html

Source reports reviewed

Perfect CashSnap Dataset - Executive Summary.pdf
Executive Summary (1).pdf
Perfect CashSnap Dataset_ Research Summary and Blueprint(1).pdf
CashSnap Dataset Strategy Research.pdf